Numerical renormalization applied to the modelling of adaptative control systems

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In this proposal, we present a novel numerical renormalization method based on the dynamical renormalization group (DRG) for modeling and controlling nonlinear dynamical systems. Our previous work demonstrated that this method offers accurate solutions without secular divergences, even for systems in which analytic solutions are unattainable. This approach is especially useful when only the linear response of the system is available. **JG: aqui tambien seria bueno que digamos algo de los casos que se explorarán**

I. INTRODUCTION

Control system engineering is a vital and evolving field that integrates efficient process design with technical forecasting skills. It has significant applications in manufacturing, transportation and environmental management. A major challenge in control system design is selecting an appropriate mathematical framework to understand how a system responds to internal and external fluctuations. Differential equations (DE) [1] often serve as the foundation for modelling complex system dynamics, providing flexibility and consistency in capturing system responses.

Exact solutions to differential equations are rarely attainable due to the difficulty of analytical calculations or the prohibitive cost of numerical solutions. Perturbation theory can help address these issues and provide approximate solutions, but many problems are not amenable to simple perturbative methods and may show singular behavior over time, requiring additional techniques to avoid uncontrolled growth. One of such techniques involves the dynamical renormalization group (DRG) [2, 3], which relies on truncated solutions up to first-order in perturbation theory to approximate solutions through reparameterizations. The DRG approach is characterized by its reparameterization based on the secular growth with time, which captures the nonlinear features of the dynamical system as the parameters evolve. However, obtaining analytic expressions for perturbative solutions is not always feasible. Thus, the DRG approach has been adapted to accommodate both numerical and analytic solutions [4]. This proposal aims to explore numerical DRG as a method for constructing solutions to nonlinear dynamical systems, focusing exclusively on their first-order perturbative behavior.

JG: Aqui entran algunos párrafos donde se describen los tipos de sistemas en los que se puede usar y el benchmarking que se busca hacer al comparar las soluciones obtenidas con otros métodos.

The layout of this paper is organized as follows: in Section II, we introduce the foundations of the DRG method and its numerical implementation in systems truncated to first-order in a perturbative expansion. To show how the method works, we produce approximate solutions of the Duffing and the Van der Pol equations as deformations of the simple harmonic oscillator. In Section III, we propose case studies relevant to nonlinear control theory and methods for benchmarking our semi-analytical results against the approximate solutions produced by other existing methods. Finally, in Section IV we discuss and conclude.

II. METHOD OF NUMERICAL RENORMALIZATION

A. Case 1: deforming the harmonic oscillator into the Duffing model

In this section, we demonstrate the generation of approximate solutions using the numerical renormalization procedure through a couple of simple examples. In our first toy model, we start with the harmonic oscillator equations

$$\frac{\mathrm{d}\varphi^{(0)}}{\mathrm{d}t} = \Pi^{(0)}\,, \quad \frac{\mathrm{d}\Pi^{(0)}}{\mathrm{d}t} = -\omega^2\varphi^{(0)}\,, \tag{1}$$

and then deform them into the Duffing equation, which is free of damping and external forces:

$$\frac{\mathrm{d}\varphi_{\mathrm{full}}}{\mathrm{d}t} = \Pi_{\mathrm{full}}\,, \tag{2a}$$

$$\frac{\mathrm{d}\Pi_{\mathrm{full}}}{\mathrm{d}t} = -\omega^2\varphi_{\mathrm{full}} - \varepsilon\varphi_{\mathrm{full}}^3\,, \tag{2b}$$

by introducing a cubic correction controlled by a tunable deformation parameter, ε. In the left panel of Fig. 1, case (a) illustrates a harmonic oscillator with spring constant k and mass m, oscillating freely around its equilibrium position. For simplicity, we consider the solution with maximum initial amplitude set to one:

$$\varphi^{(0)}(t) = \cos(\phi_\circ)\,, \quad \Pi^{(0)}(t) = -\omega\sin(\phi_\circ)\,, \tag{3}$$

where $\phi_\circ = \omega t$ is the oscillation phase and ω is the angular frequency. The oscillation phase can be calculated using

$$\frac{\mathrm{d}\phi_\circ}{\mathrm{d}t} = \omega\,, \tag{4}$$

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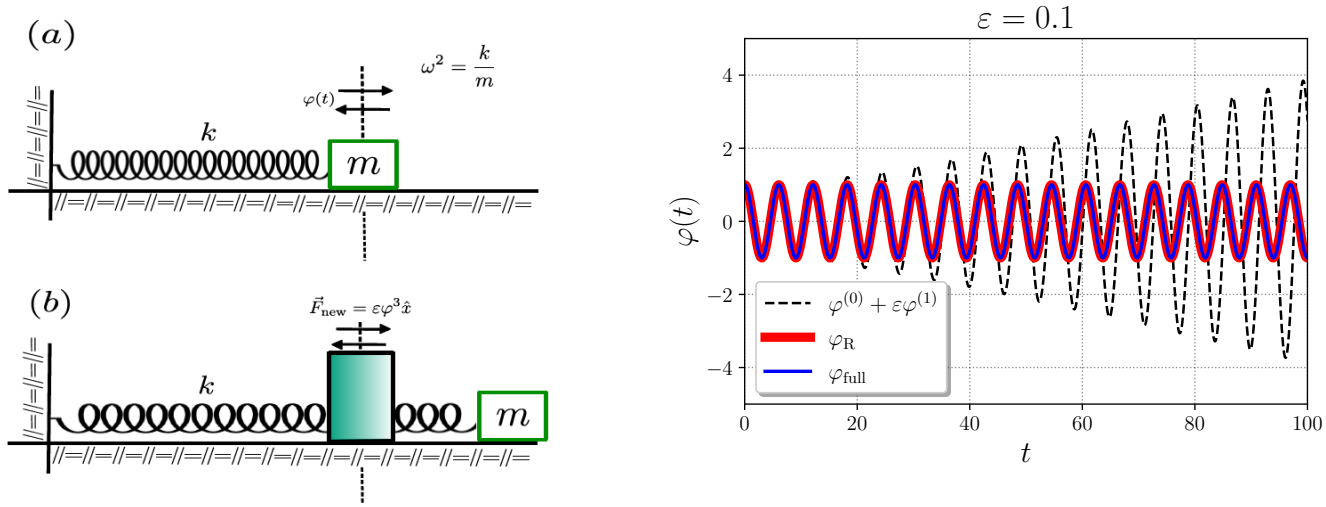


FIG. 1. Left panel: A cartoon showing the dynamical setup of a harmonic oscillator (case (a)) and its deformation by an additional force proportional to the cube of the elongation of the spring with respect to its equilibrium position. Right panel (top): Comparing the full solution ($\varphi_{\rm full}$, in a solid blue line) with the renormalized solution ($\varphi_{\rm R}$, in a solid red line) and with the solution corrected up to first order in perturbation theory ($\varphi^{(0)} + \varepsilon\varphi^{(1)}$, in a dashed black line) in the case $\varepsilon = 0.1$. The renormalized solution captures changes in the oscillation phase without introducing divergent amplitudes as it evolves.

showing that the solution is fully determined by the angular frequency.

In the lower left panel, labeled (b), Fig. 1 shows how an additional force $\vec{F}_{\rm new} = \varepsilon\varphi^3\hat{x}$ deforms the harmonic oscillator into the Duffing model. Since the equation of motion in Eq. (2b) is nonlinear, finding analytic solutions is difficult. Thus, numerical solutions are required. The thick blue curve in the right panel of Fig. 1 shows the numerical nonlinear wave solution of the Duffing oscillator with an initial amplitude of one and zero velocity in the case $\varepsilon = 0.1$.

The numerical solution provides valuable insight: the full solution is only different from the original cosine function in its oscillation phase. Therefore, it is reasonable to suggest that small corrections to $\varphi^{(0)}$ may suffice to build an approximate solution. We can introduce such small corrections perturbatively by

$$\varphi_{\rm full} \approx \varphi^{(0)} + \varepsilon\varphi^{(1)}\,, \quad \Pi_{\rm full} \approx \Pi^{(0)} + \varepsilon\Pi^{(1)} \tag{5}$$

Here, $\varphi^{(1)}$ represents a correction controlled by the deformation parameter ε. Substituting this ansatz in Eq. (2b) divides the Duffing equation into two parts: the first, proportional to ε^0, matches with equations of motion for the harmonic oscillator in Eq. (1), while the second, proportional to ε^1, gives:

$$\frac{{\rm d}\varphi^{(1)}}{{\rm d}t} = \Pi^{(1)}\,, \tag{6a}$$

$$\frac{{\rm d}\Pi^{(1)}}{{\rm d}t} = -\omega^2\varphi^{(1)} - (\varphi^{(0)})^3\,. \tag{6b}$$

Extending the prescription in Eq. (5) to higher orders in ε creates a hierarchy of equations, where higher-order corrections depend on lower orders for their solutions. This is evident in Eq. (6b), where the term $(\varphi^{(0)})^3$ acts like an external, autonomously oscillating force. By setting $\varphi^{(1)} = \Pi^{(1)} = 0$ as initial conditions, and using $\varphi^{(0)}$ from Eq. (3), the black dashed line in the right panel of Fig. 1 illustrates the solution corrected at first-order. Our results show that this first-order correction causes the amplitude to diverge after a few oscillations. This phenomenon, known as secular growth, is a well-known failure of perturbative schemes [5–8]. As shown by the numerical solutions, this divergence is an artifact of the perturbative decomposition.

In the following subsection, we explain how to reuse these first-order solutions to produce approximate solutions that are free from uncontrolled growth. Notably, this procedure is flexible in that it can take as input either analytical or numerical forms of the background and perturbative solutions.

B. Renormalized solutions from first-order perturbation theory

Intuitively, it is still valid to ask whether an approximate solution can be built from minimal modifications of $\varphi^{(0)}$. In the case of the nonlinear wave solution of the Duffing oscillator, it suggests that a time-dependent correction to the oscillation frequency and phase may be sufficient to produce a reasonable approximation. Therefore, in this section, we introduce the numerical renormalization method to determine the time-dependence of the solution parameters. Specifically, we need to account for the dependence of the solutions on a vector of background

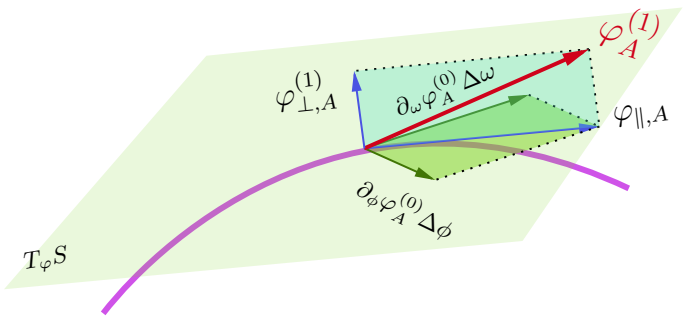


FIG. 2. Projection of the perturbative solution (vector in red, tangent to the flow of solutions in functional space) in derivatives of the background solution (in blue). First-order error is depicted as a green vector perpendicular to the tangent plane $T_\varphi S$ formed by such derivatives. The generators of parameter space transformations are the fitting coefficients minimizing the error's norm $\varphi^{(1)}_{\perp,A}$

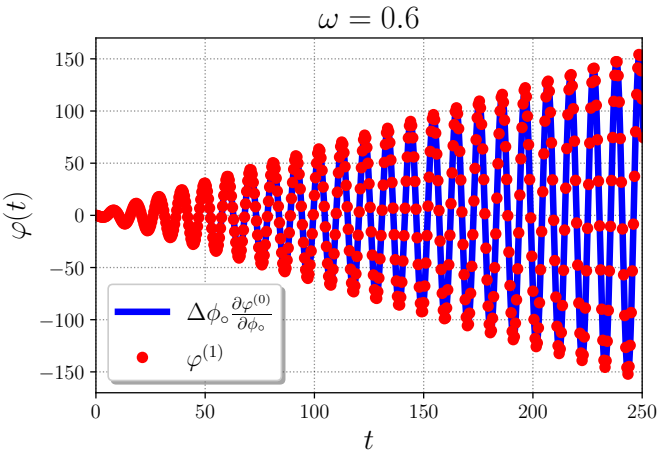


FIG. 3. Fitting the Duffing perturbation to the harmonic oscillator for $\omega = 0.6$. This plot shows that it the infinitesimal shift defined in Eq. (9b) is sufficient to capture the evolution of perturbations using background solution derivatives as a basis.

parameters, $\vec{\lambda}_0(t)$:

$$\varphi^{(0)} : \varphi^{(0)}(t;\vec{\lambda}_0)\,, \quad \vec{\lambda}_0(t) = (\phi_\circ;\omega)\,. \tag{7}$$

In the case of solutions of the original dynamical system (i.e., the harmonic oscillator), the flow of parameters with time is given by

$$\vec{\beta}^{(0)} \equiv \frac{\mathrm{d}\vec{\lambda}_0(t)}{\mathrm{d}t} = (\omega;0)\,, \tag{8}$$

which is consistent with the evolution of the phase in Eq. (4), while the angular frequency remains constant. The key idea is to deform the flow of parameters, generating both (a) time evolution and (b) reparameterizations in parameter space. This deformation is achieved through the following infinitesimal transformations ansatz:

$$\vec{\lambda}_0(t) \to \vec{\lambda}_\mathrm{R}(t) = \vec{\lambda}_0(t) + \varepsilon\Delta\vec{\lambda}\,; \tag{9a}$$

$$\Delta\vec{\lambda} \equiv (\Delta\phi_\circ;\Delta\omega) = \left(\alpha^{(1)}_\omega \Delta t; \alpha^{(1)}_\omega + \beta^{(1)}_\omega \Delta t\right)\,. \tag{9b}$$

Here, $\Delta t \equiv t - t_0$, while $\alpha^{(1)}_\omega$ governs diffeomorphisms in parameter space.[1] In what follows of this section, we will motivate the form of shifts in Eq. (9b). While these choices may seem ad hoc for this toy model, a more general method to find the generators for a broader class of problems is presented in [4].

The redefinitions in Eq. (9a) and (9b) induce $\mathrm{d}\vec{\lambda}_\mathrm{R}/\mathrm{d}t = \vec{\beta}^{(0)} + \varepsilon\vec{\beta}^{(1)}$ as a parameter flow equation, which leads to a system of coupled differential equations

$$\frac{\mathrm{d}\phi_\mathrm{R}}{\mathrm{d}t} = \omega_\mathrm{R} + \varepsilon\alpha^{(1)}_\omega(\omega_\mathrm{R})\,, \tag{10a}$$

$$\frac{\mathrm{d}\omega_\mathrm{R}}{\mathrm{d}t} = \varepsilon\beta^{(1)}_\omega(\omega_\mathrm{R})\,, \tag{10b}$$

With this parameter flow deformation, we can construct an approximate solution by plugging the new parameters into the background solution

$$\begin{aligned}\varphi_\mathrm{full} &\approx \varphi_\mathrm{R}(t) \equiv \varphi^{(0)}(t,\vec{\lambda}_\mathrm{R})\,,\\ \Pi_\mathrm{full} &\approx \Pi_\mathrm{R}(t) \equiv \Pi^{(0)}(t,\vec{\lambda}_\mathrm{R})\,,\end{aligned} \tag{11}$$

where the pair $(\varphi_\mathrm{R},\Pi_\mathrm{R})$ serves as an approximation to $(\varphi_\mathrm{full},\Pi_\mathrm{full})$. To achieve this, it is crucial to identify the generators of the parameter transformations to build such an approximate solution from the parameter flow. This identification is done by matching the solution obtained by the first-order perturbative expansion with the renormalized solution:

$$\varphi^{(0)}_A(t,\vec{\lambda}_0) + \varepsilon\varphi^{(1)}_A = \varphi_{\mathrm{R},A}(t,\vec{\lambda}_\mathrm{R}) + \varepsilon\varphi^{(1)}_{\perp,A}\,, \tag{12}$$

where the capital latin index A labels the solution ($A = 1$) and its conjugate momentum ($A = 2$). $\varphi^{(1)}_{\perp,A}$ is the estimation error at first order, which contain parts of the solutions that cannot be represented by the renormalized solution. The next step is to observe that both the parameter flow deformation in Eq. (9a) and the definition of the renormalized solution in Eq. (11) allow further expansion in ε

$$\varphi_{\mathrm{R},A}(t,\vec{\lambda}_\mathrm{R}) = \varphi^{(0)}_A(t,\vec{\lambda}_0) + \varepsilon\nabla_{\vec{\lambda}_0}\varphi^{(0)}_A\cdot\Delta\vec{\lambda}\,. \tag{13}$$

With this expansion, we collect all the terms proportional to ε, yielding

$$\varphi^{(1)}_{\perp,A} = \varphi^{(1)}_A - \varphi_{\parallel,A}\,, \tag{14}$$

[1] Under this prescription, the background flow equation $\mathrm{d}\phi_\circ/\mathrm{d}t = \omega$ remains valid for the renormalized phase, ensuring that the number of parameters required to describe the solution stays the same. Alternatively, one could assume that the flow generators deform similarly than the frequency, i.e., $\Delta\phi_\circ = \alpha^{(1)}_\phi + \beta^{(1)}_\phi\Delta t$. Although we will discuss these deformed generators, they do not affect the results derived from the flow deformations in Eq. (9b).

where

$$\varphi_{\parallel,A} \equiv \nabla_{\vec{\lambda}_0}\varphi_A^{(0)} \cdot \Delta\vec{\lambda}, . \tag{15}$$

To minimize the error's norm, we define the "cost function"

$$I(\alpha_\omega^{(1)};\beta_\omega^{(1)};\beta_\phi^{(1)}) = \sum_{A=1}^{2}\int_{t=t_0}^{\infty}|\varphi_{\perp,A}^{(1)}|^2 dt\,, \tag{16}$$

where ω varies. This minimization leads to the interpretation of the generators $(\alpha_\omega^{(1)};\beta_\omega^{(1)};\beta_\phi^{(1)})$ as projection coefficients of the perturbative solutions $\varphi_A^{(1)}$ onto the basis of functions $\nabla_{\vec{\lambda}}\varphi_A^{(0)}$. An equivalent approach is to view this process as a fitting procedure, where the time evolution and reparameterization generators "extract" the divergent behavior of the perturbative solution across a sufficiently large region of the original parameter space – corresponding to a range of angular frequencies in our toy model. Fig. 2 illustrates the geometry behind the fitting procedure: the first order error, $\varphi_{\perp,A}^{(1)}$ can be viewed as a vector perpendicular to the plane $T_\varphi S$ (shaded in light green), formed by the basis vectors $\partial\varphi_A^{(0)}/\partial\phi_\circ$ and $\partial\varphi_A^{(0)}/\partial\omega$. Minimizing the norm of the error captures the parameter flow by identifying $\beta_\phi^{(1)}$ and $\beta_\omega^{(1)}$ as fitting coefficients at each point of parameter space. Since the cost function $I(\alpha_\omega^{(1)};\beta_\omega^{(1)};\beta_\phi^{(1)})$ is a quadratic form, minimizing it with respect to the generators is a straightforward procedure.

Fig. 3 validates two aspects of the approach: (a) the choice of infinitesimal generators in Eq. (9b) to reconstruct $\varphi^{(1)}$, and (b) the minimization of $I(\alpha_\omega^{(1)};\beta_\omega^{(1)};\beta_\phi^{(1)})$ in Eq. (16), showing that this method effectively fits the Duffing perturbation $(\varphi^{(1)},\Pi^{(1)})$ as a linear combination of derivatives of harmonic oscillator's background solutions. As previously mentioned, the fitting coefficients correspond to $\Delta\phi_\circ$ and $\Delta\omega$ for each value of the angular frequency ω. This result also applies for the conjugate momentum $\Pi^{(1)}$.

After determining the generators of time evolution and initial condition reparameterizations for various values of ω, we solve the parameter flow from Eqns. (10a–10b) and construct the renormalized solutions based on Eq. (11). Our results indicate that the time generator $\beta_\omega^{(1)}$ is consistent with zero for all values of ω. This suggests that the flow equation for the oscillation phase in Eq. (8) only needs a frequency shift to hold. In general, we find that the generator of diffeomorphisms is a non-linear function of frequency. But in the specific case of $\omega = 1$, the frequency shift $\alpha_\omega = 3/8$ coincides with the multiple-scale result presented by Bender and Orszag in [7] (chapter 11, page 551). This outcome is not counterintuitive, as adding a small quartic deformation to the quadratic potential does not change drastically the system's periodic nature. The red curve in Fig. 1 shows the evolution of the renormalized solution for $\omega = 1.0$ and $\varepsilon = 0.1$, which remains in phase with the full solution for over fifty oscillations. By construction, the renormalized solution avoids uncontrolled amplitude growth.

This procedure can be applied to a wider class of problems. In reference [4], we introduced a step-by-step algorithm for constructing renormalized solutions based on a first-order perturbative scheme. This approach is amenable to dynamical systems with an arbitrary number of degrees of freedom.

C. Case 2: the Van der Pol oscillator

Thus far, our approach has relied solely on a parameter shift to generate an approximate solution of the Duffing oscillator. In this section, we will explore a case where the parameters must evolve to obtain approximate results. As a suitable example of this scenario, we present the Van der Pol oscillator [9]

$$\frac{\mathrm{d}\varphi_{\mathrm{VdP}}}{\mathrm{d}t} = \Pi_{\mathrm{VdP}}\,, \tag{17a}$$

$$\frac{\mathrm{d}\Pi_{\mathrm{VdP}}}{\mathrm{d}t} = -\omega^2\varphi_{\mathrm{VdP}} + \varepsilon(1+\varphi_{\mathrm{VdP}}^2)\Pi_{\mathrm{VdP}}\,. \tag{17b}$$

This system can also be viewed as a deformation of the harmonic oscillator, particularly in the regime of small modifications $\varepsilon \ll 1$. The numerical results in the left panel of Fig. 4 suggest that, rather than considering changes in frequency, we need to parameterize the background solutions with a varying amplitude. This leads us to represent the initial conditions for equations (17a) and (17b) as:

$$\varphi^{(0)} = \mathcal{A}_0\cos(\omega t) \quad , \quad \Pi^{(0)} = -\omega\mathcal{A}_0\sin(\omega t)\,. \tag{18}$$

By spanning over amplitudes $(\mathcal{A}_0)$ instead of frequencies, we can better capture the dynamics of the system. One might initially expect that a naive first-order perturbative approach, adding small corrections proportional to ε^1, would yield a satisfactory approximation (i.e., $\varphi_{\mathrm{VdP}} \approx \varphi^{(0)} + \varepsilon\varphi^{(1)}$). Thus, the evolution of the perturbation $\varphi^{(1)}$ and its conjugate momentum $\Pi^{(1)}$ is governed by the following equations of motion:

$$\frac{\mathrm{d}\varphi^{(1)}}{\mathrm{d}t} = \Pi^{(1)}\,, \tag{19a}$$

$$\frac{\mathrm{d}\Pi^{(1)}}{\mathrm{d}t} = -\omega^2\varphi^{(1)} + \Pi^{(0)}(1+(\varphi^{(0)})^2)\,. \tag{19b}$$

However, as with the Duffing oscillator, these first-order corrections exhibit uncontrolled amplitude growth over time. Secular growth plays a key role in determining the approximate time dependence of the amplitude. Hence, we can project the growth of perturbations onto the infinitesimally shifted parameter flow

$$\vec{\lambda}_0(t) \to \vec{\lambda}_{\mathrm{R}}(t) = \vec{\lambda}_0(t) + \varepsilon\Delta\vec{\lambda}\,, \tag{20}$$

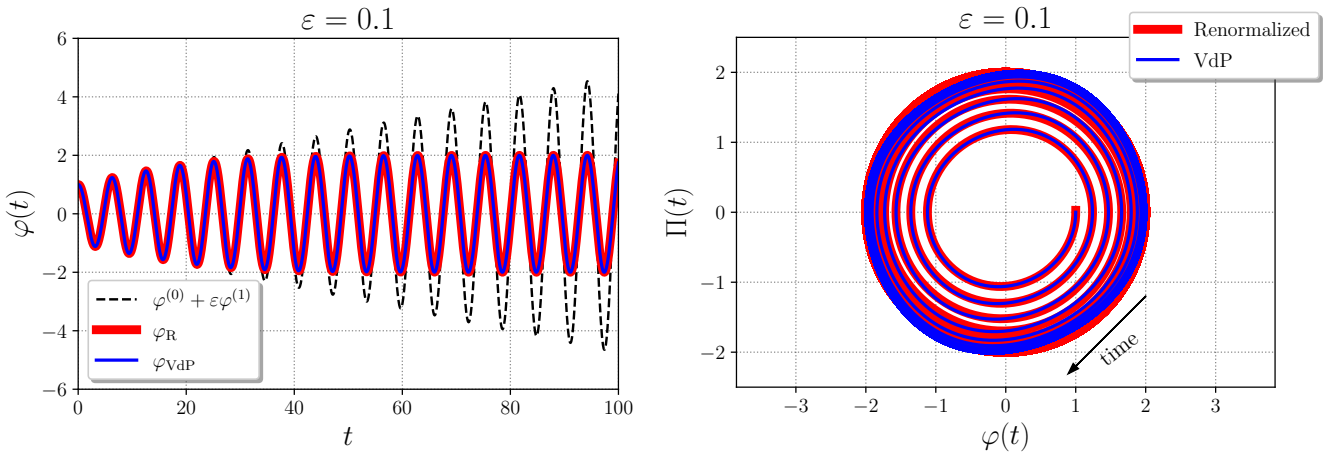


FIG. 4. Left panel: A comparison of the full solution of the Van der Pol oscillator (φ_{VdP}) with $\varepsilon = 0.1$, $\mathcal{A}_0 = 1.0$ and $\omega = 1.0$ (in blue), the renormalized solution φ_{R} (in red), and the naive perturbative solution $\varphi^{(0)} + \varepsilon\varphi^{(1)}$ (in dashed black lines). The renormalized solution closely tracks the amplitude evolution of the full solution without introducing secular growth. Right panel: Phase space plots comparing the same solutions as in the left panel. The black arrow shows the clockwise flow of the solutions toward the limit cycle, where the curves converge. As ε increases, differences in the limit cycles become clearer due to unresolved time dependence at later times.

where

$$\Delta\vec{\lambda} \equiv \Delta\mathcal{A}_0 = \beta^{(1)}_{\mathcal{A}}\Delta t\,. \tag{21}$$

In this scenario, we do not consider the generators of frequency evolution or diffeomorphisms. This leads to the following equation of motion for the renormalized amplitude

$$\frac{\mathrm{d}\mathcal{A}_{\mathrm{R}}}{\mathrm{d}t} = \varepsilon\beta^{(1)}_{\mathcal{A}}(\mathcal{A}_{\mathrm{R}})\,. \tag{22}$$

As in the previous section, we can substitute the parameter flow solution of Eq. (22) to obtain

$$\begin{aligned}\varphi_{\mathrm{VdP}} &\approx \varphi_{\mathrm{R}}(t) \equiv \varphi^{(0)}(t, \mathcal{A}_{\mathrm{R}}(t))\,, \\ \Pi_{\mathrm{VdP}} &\approx \Pi_{\mathrm{R}}(t) \equiv \Pi^{(0)}(t, \mathcal{A}_{\mathrm{R}}(t))\,,\end{aligned} \tag{23}$$

where the pair $(\varphi_{\mathrm{R}}, \Pi_{\mathrm{R}})$ serves as an approximation to φ_{full}. Following the procedure outlined in subsection II B, we find the numerical solution of equations (19a) and (19b) for every value of $\mathcal{A}_0$. From each of these solutions, we find $\beta^{(1)}_{\mathcal{A}}$ as a function of each amplitude, and derive the parameter flow by solving Eq. (22). Similar to the case of the Duffing oscillator, we find that the time-evolution generator $\beta^{(1)}_{\mathcal{A}}$ is a nonlinear function of the amplitude. Consequently, it is unsurprising that the infinitesimal amplitude ansatz, which assumes a linear time dependence of the amplitude, breaks down beyond the infinitesimal limit.

As demonstrated in the left panel of Fig. 4, much like the Duffing oscillator, the renormalized solution of the Van der Pol oscillator (φ_{R}, in solid red) effectively captures the initial amplitude growth resulting from the deformation of the cosine solution, while avoiding secular growth. However, it is important to note that long-time amplitude modulations caused by larger deformations (i.e., higher values of ε) are not adequately represented by the parameterization proposed in Eq. (18). The right panel of Fig. 4 illustrates the phase space plots of both the full and renormalized solutions, where differences between both limit cycles are visible. These differences become more pronounced as ε increases, leading to significant shape deformations.

III. CASES OF STUDY AND BENCHMARKING

IV. DISCUSSION

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